



# Applications of Linear Programming

Many real-world problems can be solved using linear programming.

## How to Solve a Linear Programming Problem

**Step 1** Define the variables.

**Step 2** Determine the objective function and whether it needs to be maximized or minimized.

**Step 3** Determine the constraints (linear inequalities).

**Step 4** Graph the constraints to construct the feasible region.

**Step 5** Find the vertex points of the feasible region.

**Step 6** Substitute the coordinates of each vertex point into the objective function to determine the maximum or minimum value of the objective function.

## EXAMPLE

That's How We Roll (THWR) manufactures two models of inline skates: Zoom and Whoosh. The number of hours required for assembling, finishing, and packaging, as well as the maximum time available for each process, are listed in the table.

| Process        | Zoom | Whoosh | Maximum |
|----------------|------|--------|---------|
| assembling (h) | 2.5  | 3.5    | 1800    |
| finishing (h)  | 0.5  | 1      | 450     |
| packaging (h)  | 2    | 2      | 1300    |

THWR makes \$20 in profit from each Zoom model sold and \$30 from each Whoosh model sold.

- A** How many of each model should THWR make to produce the maximum profit?

### SOLUTION

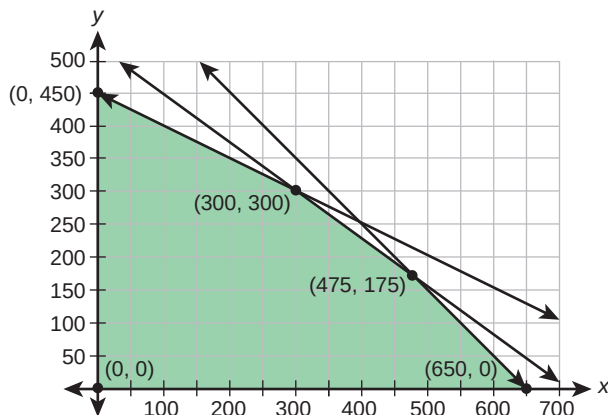
**Step 1** Let  $x$  = number of Zoom models, and let  $y$  = number of Whoosh models.

**Step 2** Since the company makes \$20 in profit from each Zoom model sold and \$30 from each Whoosh model sold, the profit function is  $P = 20x + 30y$ . The company wants to maximize profit,  $P$ .

**Step 3** Determine the constraints.

$$\begin{cases} 2.5x + 3.5y \leq 1800 & \text{no more than 1800 h assembling} \\ 0.5x + y \leq 450 & \text{no more than 450 h finishing} \\ 2x + 2y \leq 1300 & \text{no more than 1300 h packaging} \\ x \geq 0 & \text{number of Zoom models cannot be negative} \\ y \geq 0 & \text{number of Whoosh models cannot be negative} \end{cases}$$

**Step 4**



**Step 5** The vertex points of the feasible region are  $(0, 0)$ ,  $(0, 450)$ ,  $(300, 300)$ ,  $(475, 175)$ , and  $(650, 0)$ .

**Step 6** Evaluate the objective function at each vertex point.

| Vertex point | $20x + 30y$                   | $P$      |
|--------------|-------------------------------|----------|
| $(0, 0)$     | $20 \cdot 0 + 30 \cdot 0$     | \$0      |
| $(0, 450)$   | $20 \cdot 0 + 30 \cdot 450$   | \$13,500 |
| $(300, 300)$ | $20 \cdot 300 + 30 \cdot 300$ | \$15,000 |
| $(475, 175)$ | $20 \cdot 475 + 30 \cdot 175$ | \$14,750 |
| $(650, 0)$   | $20 \cdot 650 + 30 \cdot 0$   | \$13,000 |

Since the value of  $P$  is greatest at  $(300, 300)$ , the maximum profit occurs when the company sells 300 of each skate model.

**B** What is the maximum profit?

**SOLUTION**

The maximum profit is \$15,000.

**C** THWR decides to discount the price of the Whoosh by \$2 (resulting in \$28 profit from each sale). How many of each model should the company now make to maximize profits?

**SOLUTION**

Since only the objective function will change, the same feasible region and vertex points can be used in the new objective function. Since THWR now makes \$28 for each Whoosh ( $y$ ) model sold, the new objective function is  $P = 20x + 28y$ .

| Vertex point | $20x + 28y$                   | $P$      |
|--------------|-------------------------------|----------|
| $(0, 0)$     | $20 \cdot 0 + 28 \cdot 0$     | \$0      |
| $(0, 450)$   | $20 \cdot 0 + 28 \cdot 450$   | \$12,600 |
| $(300, 300)$ | $20 \cdot 300 + 28 \cdot 300$ | \$14,400 |
| $(475, 175)$ | $20 \cdot 475 + 28 \cdot 175$ | \$14,400 |
| $(650, 0)$   | $20 \cdot 650 + 28 \cdot 0$   | \$13,000 |

**D** What is the adjusted maximum profit?

**SOLUTION**

The adjusted maximum profit is \$14,400. ■